You are a quantitative finance researcher.

I am building a trading strategy for a portfolio composed only of Bitcoin (BTC) and Ethereum (ETH).

Provide a structured academic overview of the main strategy families for a two-asset crypto portfolio, including:

Strategic allocation / buy & hold

Rebalancing (calendar, threshold, volatility-scaled)

Momentum / trend following

Mean reversion / relative value (BTC-ETH spread)

Volatility harvesting

Risk parity / volatility targeting

Liquidity provision / market making

Derivatives overlays (basis, funding, options)

Regime-switching and adaptive models

For each strategy:

Core intuition and basic mathematical formulation

Required signals/data

Key assumptions about BTC-ETH dynamics

Typical pitfalls and failure modes (overfitting, regime breaks, costs, tail risk)

Also summarize stylized empirical facts about BTC-ETH correlation, spread behavior, and volatility regimes.

Use an academic tone and focus on mechanisms rather than heuristics.

Stylized BTC–ETH joint dynamics

Empirical work on BTC and ETH documents several robust features relevant for all strategy families:

- **High, time-varying correlation and spillovers.** BTC–ETH correlations are generally positive and increase in stress regimes such as COVID-19; VAR-AGARCH and BEKK models find significant return and volatility spillovers between BTC and ETH, with higher conditional correlations and hedge ratios in crisis periods (Yousaf & Ali, 2020; Katsiampa, 2018; Gong & Huser, 2019; Cremaschini et al., 2022). Tail dependence, especially in the **lower tail**, has strengthened over time: BTC–ETH moved from near tail-independence to pronounced joint crashes (Gong & Huser, 2019).
- **Volatility clustering, heavy tails, structural breaks.** Both assets show leptokurtic returns, volatility clustering, and regime shifts in volatility (Katsiampa, 2018; Zhang et al., 2018; Cremaschini et al., 2022). GARCH-type models (including Student-t, EGARCH, and DCC-AGARCH) fit BTC/ETH volatility well, and regime-switching or threshold mechanisms improve tail-risk forecasts (Katsiampa, 2018; Daniel & Josaphat, 2025; Lee, 2025; Chen et al., 2025; Gao et al., 2022; Haidar et al., 2025).
- **Asymmetric, regime-dependent risk.** Volatility and correlations respond asymmetrically to negative shocks, with stronger effects in high-volatility regimes, and crisis periods show more expensive hedging and higher hedge effectiveness (Yousaf & Ali, 2020; Katsiampa, 2018; Karim et al., 2024; Bruzgė et al., 2023; Yakymashko, 2025).
- **Bitcoin dominance.** Optimal portfolios against broad crypto indices tend to overweight BTC relative to ETH, reflecting its lower relative risk and "store-of-value" role (Yousaf & Ali, 2020; Sanhaji & Chevallier, 2023).

These properties imply: (i) diversification between BTC and ETH is real but limited in crises; (ii) spread and volatility dynamics are regime-dependent and heavy-tailed; (iii) any model assuming constant correlation, Gaussian spreads, or stationary volatility is fragile.

BTC–ETH correlation, spread, and regimes (summary table)

Feature	Stylized behavior	Implication for strategies	Citations
Linear correlation	Positive, time-varying; higher in crises	Rebalancing & RV benefits shrink in stress	(Yousaf & Ali, 2020; Katsiampa, 2018; Gong & Huser, 2019; Cremaschini et al., 2022)
Tail dependence	Strengthening lower-tail link, moderate upper tail	Joint crashes; tail-risk critical	(Gong & Huser, 2019; Haidar et al., 2025)
Volatility	Clustered, persistent, regime-switching, heavy-tailed	Vol-targeting, risk parity, options overlays natural	(Katsiampa, 2018; Zhang et al., 2018; Daniel & Josaphat, 2025; Lee, 2025)
Spillovers	Bidirectional return & vol spillovers	BTC shocks quickly transmit to ETH	(Yousaf & Ali, 2020; Katsiampa, 2018; Yakymashko, 2025)

FIGURE 1 Key BTC–ETH joint dynamics relevant for portfolio design.

1. Strategic allocation / buy-and-hold

Intuition and formulation. Choose fixed weights (w_{BTC}, w_{ETH}) to maximize expected utility or mean–variance objective over horizon (T):

$$\max_{w} ; \mathbb{E}[R_p] - \frac{\gamma}{2}\mathrm{Var}(R_p), \quad R_p = w^\top r,$$

subject to $(w_{BTC}+w_{ETH}=1)$. In practice, BTC is often optimally overweighted relative to ETH for risk–return trade-offs (Yousaf & Ali, 2020; Sanhaji & Chevallier, 2023).

Required data. Daily (or higher-frequency) returns; estimates of means, variances, and covariance; possibly downside-risk (VaR/ES) to reflect heavy tails (Bruzgė et al., 2023; Haidar et al., 2025).

Key assumptions.

- Stationary (or slowly varying) moments of BTC and ETH returns.
- Stable correlation structure and no structural breaks.
- Investor is indifferent to path-dependence and tail clustering beyond variance.

Pitfalls.

- **Regime breaks**: Volatility regimes and tail dependence shift abruptly (Katsiampa, 2018; Gao et al., 2022; Cremaschini et al., 2022); static weights can be suboptimal or dangerous.
- **Underestimation of tails** if Gaussian or constant-volatility assumptions are used (Zhang et al., 2018; Lee, 2025; Haidar et al., 2025).
- Concentration risk: BTC dominance can be beneficial but magnifies joint crash exposure (Sanhaji & Chevallier, 2023; Gong & Huser, 2019).

---

2. Rebalancing (calendar, threshold, volatility-scaled)

Intuition and formulation.

- **Calendar rebalancing.** At fixed times (t_k), reset portfolio weights to target (w^*). Portfolio dynamics:

$$w_{t_{k}^+} = w^*, \quad w_{t} \text{ drifts with relative returns for } t\in(t_k,t_{k+1}).$$

- **Threshold rebalancing.** Rebalance when $(|w_{BTC,t}-w^*_{BTC}|>\delta)$.
- **Volatility-scaled rebalancing.** Adjust target weights or leverage using forecast vol, e.g.

$$w_t \propto \frac{1}{\hat{\sigma}t}, \quad \hat{\sigma}t^2 = \omega+\alpha r{t-1}^2+\beta \hat{\sigma}{t-1}^2,$$

using GARCH-type forecasts (Katsiampa, 2018; Chi & Hao, 2021; Daniel & Josaphat, 2025; Lee, 2025).

Required data. High-frequency or daily prices; transaction cost estimates; volatility forecasts (GARCH, realized volatility, EWMA) (Katsiampa, 2018; Chi & Hao, 2021; Daniel & Josaphat, 2025; Lee, 2025; Liu et al., 2020).

Assumptions.

- Reversion of relative weights (implicit contrarian bet).
- Volatility forecasts are sufficiently stable and predictive in presence of clustering (Katsiampa, 2018; Daniel & Josaphat, 2025; Lee, 2025).
- Correlation structure not collapsing to 1 in crises (otherwise rebalancing is just leverage management).

Pitfalls.

- **Negative convexity in trends.** Mechanical rebalancing sells winners and buys losers; in strong trends this raises drawdown risk (Rattray et al., 2019). Crypto's pronounced trends amplify this effect.
- **Transaction costs and slippage**, especially for threshold rules in high volatility.
- **Model risk in vol-scaling**: misspecified GARCH or regime-switching can lead to procyclical deleveraging or forced re-risking at wrong times (Katsiampa, 2018; Chen et al., 2025; Gao et al., 2022; Lee, 2025).
- Correlation spikes in crises reduce diversification benefits, making rebalancing less effective (Yousaf & Ali, 2020; Gong & Huser, 2019; Cremaschini et al., 2022).

---

3. Momentum / trend following

Intuition and formulation. Exploit serial dependence and persistent trends in BTC and ETH. Simple time-series momentum:

$$s_{i,t} = \mathrm{sign}\Big(\sum_{k=1}^{L} r_{i,t-k}\Big), \quad w_{i,t} \propto s_{i,t},$$

with optional vol-scaling or cross-sectional ranking between BTC vs ETH.

Required data. Return histories; realized or GARCH vol for risk-scaling; possibly macro/sentiment indicators that co-move with volatility regimes (Yakymashko, 2025; Karim et al., 2024; Lee, 2025).

Assumptions.

- Presence of trend or autocorrelation beyond noise, linked to volatility regimes and speculative dynamics (Katsiampa, 2018; Zhang et al., 2018; Cremaschini et al., 2022).
- Trend behavior broadly stable across regimes (though amplitude may vary).

Pitfalls.

- **Regime shifts** from trending to mean-reverting markets cause momentum crashes. Crypto markets show structural breaks tied to macro and regulatory news (Katsiampa, 2018; Yakymashko, 2025; Gao et al., 2022; Cremaschini et al., 2022).
- **Crowding and slippage** in highly correlated BTC–ETH up- or down-moves.
- Volatility clustering leads to **time-varying Sharpe**; static thresholds can overtrade in noise.

---

4. Mean reversion / relative value (BTC–ETH spread)

Intuition and formulation. Assume a stationary BTC–ETH relationship, e.g., log-price spread

$$X_t = \log P_{ETH,t} - \beta \log P_{BTC,t},$$

following a mean-reverting process, such as Ornstein–Uhlenbeck:

$$dX_t = \kappa(\mu - X_t)dt + \sigma dW_t,$$

possibly with regime-switching and fat-tailed innovations (García-Risueño et al., 2025; Gong & Huser, 2019). Go long ETH / short BTC when X_t is below μ, and the reverse when above.

Required data. Synchronized BTC and ETH prices; estimation of spread cointegration (if any), OU parameters, and residual distribution tails (García-Risueño et al., 2025; Gong & Huser, 2019; Cremaschini et al., 2022).

Assumptions.

- Long-run equilibrium or cointegrated relation between BTC and ETH.
- Mean-reversion speed and equilibrium level stable, or at least regime-switching but estimable.
- Spread shocks not dominated by structural breaks.

Pitfalls.

- **Fat-tailed spread residuals and jumps.** Optimal entry/exit thresholds are sensitive to heavy tails and regime-switching (García-Risueño et al., 2025; Gong & Huser, 2019).
- **Regime breaks** in technology or market structure (e.g., ETH 2.0, ETF approvals) can permanently shift the spread (Yakymashko, 2025; Cremaschini et al., 2022).
- Strengthening lower-tail dependence (Gong & Huser, 2019)increases probability of joint crashes rather than pure relative-value dislocations.
- Funding and shorting constraints in derivatives markets can impair execution.

---

5. Volatility harvesting

Intuition and formulation. For two positively correlated, volatile assets with rebalancing, convexity of log-wealth can produce a “rebalancing premium”:

$$\mathbb{E}[\Delta \log V] \approx \bar{\mu} - \frac{1}{2}\bar{\sigma}^2 + \text{diversification / rebalancing term},$$

where the extra term depends on volatilities and correlation. Crypto’s high vol and non-perfect correlation make this effect potentially large (Firouzi, 2025; Yousaf & Ali, 2020; Katsiampa, 2018; Zhang et al., 2018).

Required data. Joint return distribution, time-varying covariance estimates (rolling or GARCH/DCC) to quantify realized “volatility pumping” (Firouzi, 2025; Katsiampa, 2018; Gong & Huser, 2019; Cremaschini et al., 2022).

Assumptions.

- Sufficient volatility with less-than-perfect correlation.
- No permanent divergence of expected returns that would make one asset strictly dominated.
- Stationary covariance structure over the harvesting horizon.

Pitfalls.

- In stress regimes, correlations rise and lower-tail dependence increases, reducing the volatility-harvesting benefit exactly when risk is highest (Yousaf & Ali, 2020; Gong & Huser, 2019).
- Requires **frequent rebalancing**, raising costs and slippage; benefits can be fully offset by fees/spreads in crypto venues.
- Assumes **log-normal-like** behavior; extreme fat tails and jumps (common in BTC/ETH) undermine Taylor-series approximations (Katsiampa, 2018; Zhang et al., 2018; Lee, 2025).

---

6. Risk parity / volatility targeting

Intuition and formulation.

- **Risk parity (two-asset).** Choose weights so each asset contributes equally to portfolio volatility:

$$w_{i,t} \propto \frac{1}{\sigma_{i,t}}, \quad \text{with} \quad RC_{i,t} = w_{i,t} (\Sigma_t w_t)_i \text{ equalized},$$

using time-varying covariance Σ_t (Firouzi, 2025; Katsiampa, 2018; Chi & Hao, 2021).

- **Volatility targeting.** Scale overall exposure L_t to hit target portfolio vol σ^*:

$$L_t = \frac{\sigma^*}{\hat{\sigma}_{p,t}}, \quad w_t^{eff} = L_t w_t,$$

where $\hat{\sigma}_{p,t}$ from GARCH, realized vol, or RiskMetrics-type models (Katsiampa, 2018; Chi & Hao, 2021; Daniel & Josaphat, 2025; Liu et al., 2020; Lee, 2025).

Required data. Rolling or model-based vol/covariance estimates; intraday data can improve realized volatility and beta estimation (Katsiampa, 2018; Karim et al., 2024; Sanhaji & Chevallier, 2023; Lee, 2025).

Assumptions.

- Volatility forecasts are informative and relatively robust to model misspecification (Chi & Hao, 2021; Daniel & Josaphat, 2025; Lee, 2025).
- Leverage and de-risking can be implemented cheaply and continuously.
- Correlations not degenerate (to preserve risk-parity meaning).

Pitfalls.

- **Procyclicality:** in sharp volatility spikes, deleveraging can lock in losses and force selling into crashes (Chen et al., 2025; Gao et al., 2022; Haidar et al., 2025).
- GARCH and related models are sensitive to structural breaks and fat tails; threshold or regime-switching models improve but complicate implementation (Katsiampa, 2018; Chen et al., 2025; Gao et al., 2022; Lee, 2025).
- In a two-asset crypto-only universe with strong co-moves, “risk parity” can collapse to similar weights; tail-risk parity is not guaranteed.

---

7. Liquidity provision / market making

Intuition and formulation. Earn the bid-ask spread and liquidity premia by posting buy and sell quotes in BTC and ETH spot or perpetuals, often hedged to small net inventory. Mathematically, inventory-based market-making models (e.g., Avellaneda–Stoikov) adjust quotes based on inventory (q_t), midprice (S_t), and volatility (σ_t):

$$\delta^{bid/ask}_t = \frac{\gamma \sigma_t^2 T}{2} \pm \frac{1}{\gamma} \ln\left(1+\frac{\gamma}{k} q_t\right),$$

extended to a vector of assets and cross-asset inventory.

Required data. High-frequency order book, trade prints, realized vol, microstructure noise characteristics; cross-asset volatility and jump behavior (Katsiampa, 2018; Yakymashko, 2025; Karim et al., 2024; Sanhaji & Chevallier, 2023).

Assumptions.

- Sufficient order-flow predictability and spread stability to earn positive expected P&L net of adverse selection.
- Market microstructure remains comparable across regimes.

Pitfalls.

- Crypto markets show **large jumps and volatility spikes**, with intraday jump contributions up to ~25% in BTC/ETH (Sanhaji & Chevallier, 2023); inventories can be severely hit.
- Correlated crashes and strengthening lower-tail dependence (Gong & Huser, 2019)create cross-asset inventory risk (BTC and ETH both gap against maker).
- Exchange microstructure risk: outages, auto-deleveraging, funding liquidations.

---

8. Derivatives overlays (basis, funding, options)

Intuition and formulation.

- **Basis/futures:** Trade the spread between futures price (F_t) and spot (S_t) (cash-and-carry or reverse).
- **Funding-rate arbitrage:** On perpetual swaps, take positions when funding deviates from expected fair level.
- **Options overlays:** Use options to shape portfolio payoff and volatility exposure, e.g., selling volatility when GARCH-implied vol exceeds option-implied vol (Chi & Hao, 2021; Karim et al., 2024).

Simple options mispricing strategy:

$$\text{Signal}_t = \hat{\sigma}^{GARCH}_{t+1} - \sigma^{IV}_t,$$

short options when signal ($\gg 0$), delta-hedged (Chi & Hao, 2021).

Required data. Futures and options prices; implied volatility surfaces (BitVol, EthVol); realized and model-implied volatility series (Chi & Hao, 2021; Karim et al., 2024; Lee, 2025).

Assumptions.

- Volatility and basis are forecastable; GARCH or realized vol can predict future realized vol better than implied (Chi & Hao, 2021; Karim et al., 2024; Lee, 2025).
- Margin and funding constraints manageable, and counterparty risk controlled.

Pitfalls.

- **Tail risk from short-vol or basis trades** in a market with extreme jumps and clustering (Katsiampa, 2018; Chi & Hao, 2021; Chen et al., 2025; Lee, 2025; Haidar et al., 2025).
- Structural regime changes in implied vs realized volatility relationships; model underperformance out of sample (multi-variate volatility models underperform simple GARCH in out-of-sample crypto options forecasting (Chi & Hao, 2021)).
- Liquidity concentration in BTC vs ETH options can distort spreads and execution quality.

---

9. Regime-switching and adaptive models

Intuition and formulation. Explicitly model discrete market regimes (e.g., low-vol vs high-vol; bubble vs normal) and adapt strategy parameters or allocations accordingly.

Generic Markov-switching framework:

[r_t \mid s_t \sim \mathcal{N}(\mu_{s_t}, \Sigma_{s_t}), \quad \mathbb{P}(s_t = j \mid s_{t-1}=i) = p_{ij},]

or regime-switching CAViaR/ES-CAViaR for tail risk:

[\text{VaR}*t^{(s_t)} = f\big(\text{VaR}*{t-1}^{(s_t)},, r_{t-1};, \theta_{s_t}\big), \quad s_t \text{ Markovian},]

possibly driven by bubble or macro indicators (Chen et al., 2025; Gao et al., 2022; Cremaschini et al., 2022). Trading rules (e.g., leverage, rebalancing frequency, momentum lookback) are conditioned on (s_t).

Required data. Returns and realized volatility; macro, sentiment, or bubble indices; estimation via hidden Markov models or threshold rules (Yakymashko, 2025; Chen et al., 2025; Gao et al., 2022; Cremaschini et al., 2022; Yakymashko, 2025).

Assumptions.

- Regimes are persistent and recurrent, with relatively stable transition dynamics.
- Sufficient data per regime to estimate parameters in presence of heavy tails.
- Observable proxies (e.g., volatility level, bubble index, macro announcements) correlate strongly with latent regimes (Yakymashko, 2025; Gao et al., 2022; Chen et al., 2025).

Pitfalls.

- **Overfitting**: high-dimensional regime models on short crypto histories risk spurious regimes.
- **Regime misclassification** near switching points; strategies can whipsaw exactly at turning points, especially when volatility jumps (Katsiampa, 2018; Sanhaji & Chevallier, 2023; Haidar et al., 2025).
- Tail behavior can differ across regimes and be under-captured unless explicitly modeled with fat-tailed or threshold structures (García-Risueño et al., 2025; Chen et al., 2025; Lee, 2025; Haidar et al., 2025).

---

10. Cross-strategy considerations and failure modes

Across all families, BTC–ETH specific risks include:

- **Heavy-tailed, clustered volatility and jumps**, requiring robust volatility and tail-risk modeling (GARCH variants, ES-CAViaR, copulas for tail dependence) (Katsiampa, 2018; Chi & Hao, 2021; Chen et al., 2025; Daniel & Josaphat, 2025; Zhang et al., 2018; Lee, 2025; Haidar et al., 2025).
- **Time-varying and asymmetric dependence**, especially stronger co-crashes in recent years (Yousaf & Ali, 2020; Katsiampa, 2018; Gong & Huser, 2019; Cremaschini et al., 2022; Haidar et al., 2025).
- **Structural breaks and crises**, where hedging costs rise yet hedging effectiveness also increases, indicating that diversification properties change nonlinearly (Yousaf & Ali, 2020; Katsiampa, 2018; Yakymashko, 2025; Gao et al., 2022).

Mechanism-focused design therefore favors: explicit modeling of volatility and regimes; robust spread and tail modeling; careful leverage and rebalancing rules that temper negative convexity; and derivatives overlays that acknowledge persistent mispricings while controlling for extreme tail scenarios.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

Firouzi, K. (2025). Quantifying Crypto Portfolio Risk: A Simulation-Based Framework Integrating Volatility, Hedging, Contagion, and Monte Carlo Modeling. **.

Yousaf, I., & Ali, S. (2020). Discovering interlinkages between major cryptocurrencies using high-frequency data: new evidence from COVID-19 pandemic. *Financial Innovation*, 6. https://doi.org/10.1186/s40854-020-00213-1

Katsiampa, P. (2018). An Empirical Investigation of Volatility Dynamics in the Cryptocurrency Market. *Capital Markets: Asset Pricing & Valuation eJournal*. https://doi.org/10.2139/ssrn.3202317

Chi, Y., & Hao, W. (2021). Volatility models for cryptocurrencies and applications in the options market. *Journal of International Financial Markets, Institutions and Money*. https://doi.org/10.1016/j.intfin.2021.101421

García-Risueño, P., Ortas, E., & Moneva, J. (2025). The Effect of Fat Tails on Rules for Optimal Pairs Trading: Performance Implications of Regime Switching with Poisson Events. *International Journal of Financial Studies*. https://doi.org/10.3390/ijfs13020096

Yakymashko, V. (2025). Volatility Clustering and Market Sentiment: A Quantitative Assessment of Bitcoin and Ethereum's Reaction to Macroeconomic Announcements.. *The American Journal of Management and Economics Innovations*. https://doi.org/10.37547/tajmei/volume07issue07-07

Chen, C., Chen, P., & Hsu, Y. (2025). Bayesian Forecasting of Value-at-Risk and Expected Shortfall in Cryptocurrency Markets: A Nonlinear Semi-Parametric Framework. *Applied Stochastic Models in Business and Industry*. https://doi.org/10.1002/asmb.2926

Daniel, A., & Josaphat, A. (2025). Modeling Volatility Using Bayesian GARCH with Student-t and Generalized Error Distributions: A Case Study of Bitcoin. *Asian Journal of Science, Technology, Engineering, and Art*. https://doi.org/10.58578/ajstea.v3i4.5926

Karim, M., Rasid, M., Noman, A., & Yarovaya, L. (2024). Exploring asymmetries in cryptocurrency intraday returns and implied volatility: New evidence for high-frequency traders. *International Review of Financial Analysis*. https://doi.org/10.1016/j.irfa.2024.103617

Gao, L., Ye, W., & Guo, R. (2022). Jointly forecasting the value-at-risk and expected shortfall of Bitcoin with a regime-switching CAViaR model. *Finance Research Letters*. https://doi.org/10.1016/j.frl.2022.102826

Bruzgė, R., Černevičienė, J., Šapkauskienė, A., Mačerinskienė, A., Masteika, S., & Driaunys, K. (2023). STYLIZED FACTS, VOLATILITY DYNAMICS AND RISK MEASURES OF CRYPTOCURRENCIES. *Journal of Business Economics and Management*. https://doi.org/10.3846/jbem.2023.19118

Sanhaji, B., & Chevallier, J. (2023). Tracking 'Pure' Systematic Risk with Realized Betas for Bitcoin and Ethereum. *Econometrics*. https://doi.org/10.3390/econometrics11030019

Rattray, S., Granger, N., Harvey, C., & Hemert, O. (2019). Strategic Rebalancing. **, 46, 10 - 31. https://doi.org/10.2139/ssrn.3330134

Liu, W., Semeyutin, A., Lau, C., & Gozgor, G. (2020). Forecasting Value-at-Risk of Cryptocurrencies with RiskMetrics type models. *Research in International Business and Finance*, 54, 101259. https://doi.org/10.1016/j.ribaf.2020.101259

Zhang, W., Wang, P., Li, X., & Shen, D. (2018). Some stylized facts of the cryptocurrency market. *Applied Economics*, 50, 5950 - 5965. https://doi.org/10.1080/00036846.2018.1488076

Gong, Y., & Huser, R. (2019). Asymmetric tail dependence modeling, with application to cryptocurrency market data. *The Annals of Applied Statistics*. https://doi.org/10.1214/21-aoas1568

Cremaschini, A., Punzo, A., Martellucci, E., & Maruotti, A. (2022). On stylized facts of cryptocurrencies returns and their relationship with other assets, with a focus on the impact of COVID-19. *Applied Economics*, 55, 3675 - 3688. https://doi.org/10.1080/00036846.2022.2117777

Lee, M. (2025). Temporal Fusion Transformer-Based Trading Strategy for Multi-Crypto Assets Using On-Chain and Technical Indicators. *Syst.*, 13, 474. https://doi.org/10.3390/systems13060474

Haidar, A., Aziz, F., Khan, I., Ashfaque, M., & Hussain, A. (2025). The Strategic Role of Bitcoin in Asset Allocation: Implications for Risk Management and Long-Term Investment Performance. *Journal of Management & Social Science*. https://doi.org/10.63075/2at5qa32